Note: Write all the required equations (without simplification) to solve the following problems, then find the final answer.

Prob. 1: ***Consider the circuit shown in Fig. 1,***

1. The open circuit voltage between *a-b* terminals (V_o), when the load impedance is disconnected, is:

2. The short circuit current (I_{sc}), when *a-b* terminals are shorted is:

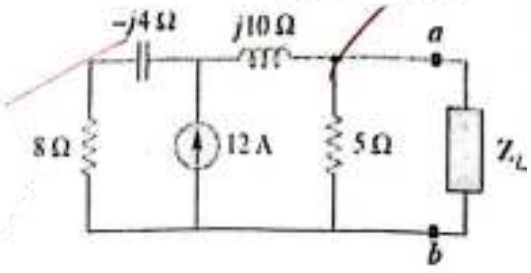


Fig.1

3. If the load impedance $Z_L = 4+j3\,\Omega$, The load current is:

Prob. 2: ***Consider the circuit shown in Fig. 2 and using mesh analysis method,***

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_c, is: .

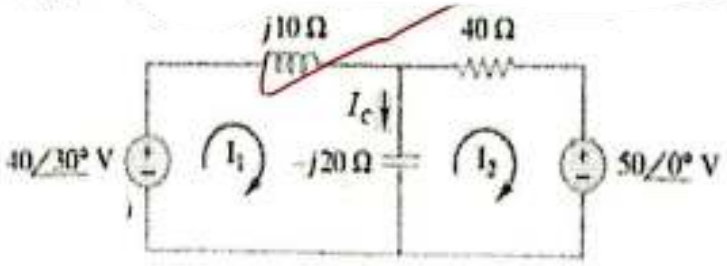


Fig 2

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Prob.3:
Consider the circuit shown in figure 3, where a voltage source $v_s(t) = 50\cos(500t + 15)\,V$ supplies the load by a current $i(t) = 5\cos(500t - 30^0)\,A$

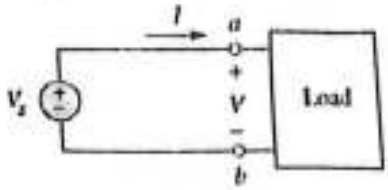


Fig.3

1. The equivalent impedance at the terminals of the voltage source is $Z =$ Ω
2. The voltage lags the current by ..... degrees.
3. The current leads the voltage by .... radians.
4. The voltage value $v_s(t)$ at $t = 2ms$ is V
5. The phasor domain representation of $v(t)$ is ..V
6. The frequency of the voltage $v_s(t)$ in Hz is .Hz
7. The average value of the voltage $v_s(t)$ is V
8. The effective value of the current $i_s(t)$ is .....A
9. Draw the phasor diagram for the voltage $v_s(t)$ and the current $i_s(t)$.

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